

Reader Head Protocol Summary

Chris, 12-Jan-2004

Standard Message Format

This applies to both *Control Unit to Reader Head* and *Reader Head to Control Unit* messages.

<SOH><address><STX><message><ETX><checksum><EOT>

Reply

<SOH><address><ACK> If checksum OK.

>

<SOH><address><NAK> If checksum wrong.

>

Note that a *Standard Message* cannot be sent as a reply.

Addresses

The Reader Head needs to respond to *Broadcast* as well as *ADH*. Need to check what the 4 button keyswitch uses.

99 Broadcast.

01 Payout keyset.

02 Reader Head (ADH).

03 Wining Number Display (WND).

04 Auto Key Switch? (AKS).

Checksum

Checksum is sum of all characters from <SOH> to <ETX> inclusive, ANDed with 0x3F, then ORed with 0x40, to make it a printable character.

e.g. <SOH>02<STX>02N<ETX> adds up to 0x118, which ANDed and Ored gives 0x58 (X), therefore the full message is <SOH>02<STX>02N<ETX>X<EOT>.

Polling Message

<ENQ><02><EOT> Control Unit to Reader Head (02 is the ADH address – see above).

Reply

<ETB> If no data to send, or
if data available.

Standard Message

Messages: Control Unit to Reader Head

These can be sent at any time, and the Reader Head must reply with <SOH><02><ACK>. Note commands are case sensitive.

<SOH><02><STX><1p><ETX><checksum><EOT>
<SOH><02><STX><2p><ETX><checksum><EOT>
<SOH><02><STX><1a><ETX><checksum><EOT>
<SOH><02><STX><0a><ETX><checksum><EOT>
<SOH><02><STX><xxN><ETX><checksum><EOT>

Set 1 pass mode.

Set 2 pass mode.

Power-up system (not supported by ARH-2).

Power-down system (not supported by ARH-2).

NMB setting. **xx** = 01 to 20

(not normally used – generally use the ARH-2 hex switch).

<SOH><02><STX><xt><ETX><checksum><EOT>
<SOH><02><STX><l><ETX><checksum><EOT>
<SOH><02><STX><4s><ETX><checksum><EOT>

Obsolete (NMB setting, **x** = 1 to 4).

Obsolete (calibrate camera based ARH).

Obsolete (spin counter). Still sent by Power Poller.

Messages: Reader head to Control Unit

These can only be sent in response to a *Polling Message*. Note commands are case sensitive.

<SOH><02><STX><wwRcDssV><ETX><checksum>
<EOT>

Winning Number, e.g. 25R1D33V.

ww = Winning number, 00 to 37, where 37=double zero

c = 1 for clockwise or 0 for anticlockwise

ss = speed in RPM

<SOH><02><STX><E><ETX><checksum><EOT>
<SOH><02><STX><G><ETX><checksum><EOT>
<SOH><02><STX><xxN><ETX><checksum><EOT>
<SOH><02><STX><S0><ETX><checksum><EOT>

Empty wheel.

Good Luck.

No more bets (**xx** is a random number).

Ball pass / sync. In principle can send S1, S2 etc for each pass, but ARH-2 only implements S0.

Some TCS documentation refers to 0S as the format, but we suspect this is a typo.

<SOH><02><STX><ZzRcDss><ETX><checksum><EOT>

Wheel data (only implemented in a special Novamatics SW version), e.g. Z1R0D05V.

z = 1 for a single zero wheel or 0 for a double zero wheel

c = 1 for clockwise or 0 for anticlockwise

ss = speed in RPM